

Thermoelectric Power Plant Watershed Stress Classification

A Quantile Regression & Spatial Risk Approach

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The Hook & The Pivot

- **The Big Problem:** Aggregate statistics mask localized extremes. Thermoelectric power withdraws ~40% of U.S. freshwater, but "average" grid metrics fail to identify specific facilities stressing vulnerable watersheds.
- **Our Original Goal:** Predict Data Center Water Usage Effectiveness (WUE) to identify inefficient facilities.
- **The Roadblock:** Severe corporate data opacity. Sustainability reports only provide portfolio-level averages, yielding insufficient sample sizes for statistical learning.
- **The Pivot:** We shifted to thermoelectric power plants—a thermodynamic analogue to data centers—utilizing highly standardized, public panel data (EIA, eGRID, NOAA) to build a robust covariate backbone.

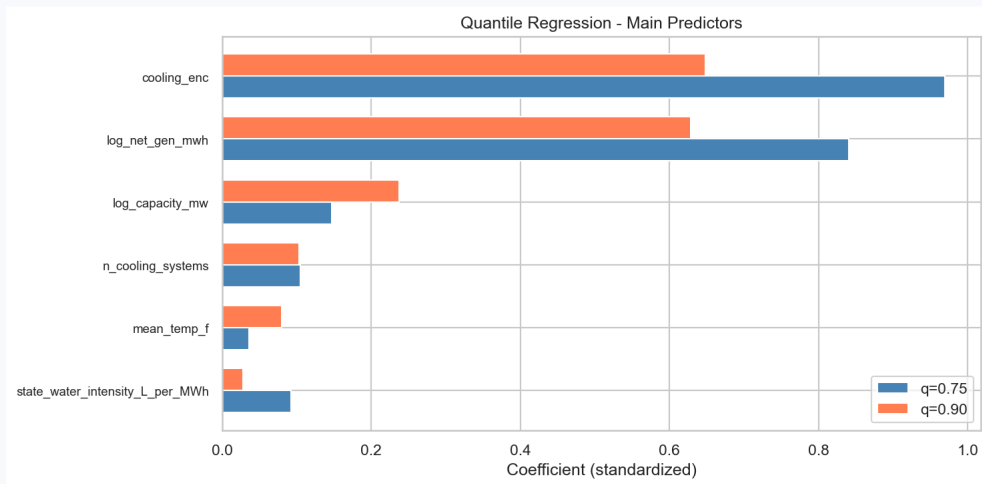
Feature Engineering & LASSO

- **The Dataset:** A longitudinal panel of 2,248 plant-year observations (2021–2024) across 46 states.
- **Target Variable:** Total Water Consumption, log-transformed ($\log(1+x)$) to account for heavy right-skewness in plant sizes.
- **The Method:** Applied LASSO (L1 regularization) with 5-fold cross-validation ($\alpha=0.058$).
- **The Result:** The L1 penalty successfully forced 24 noisy variables to zero, leaving a sparse, interpretable model with 7 dominant predictors.
- **The Predictors:** Net Generation MWh, Capacity MW, Number Cooling Systems, Cooling Category, Primary Fuel, State Water Intensity L/MWh

Establishing a Consumption Ceiling

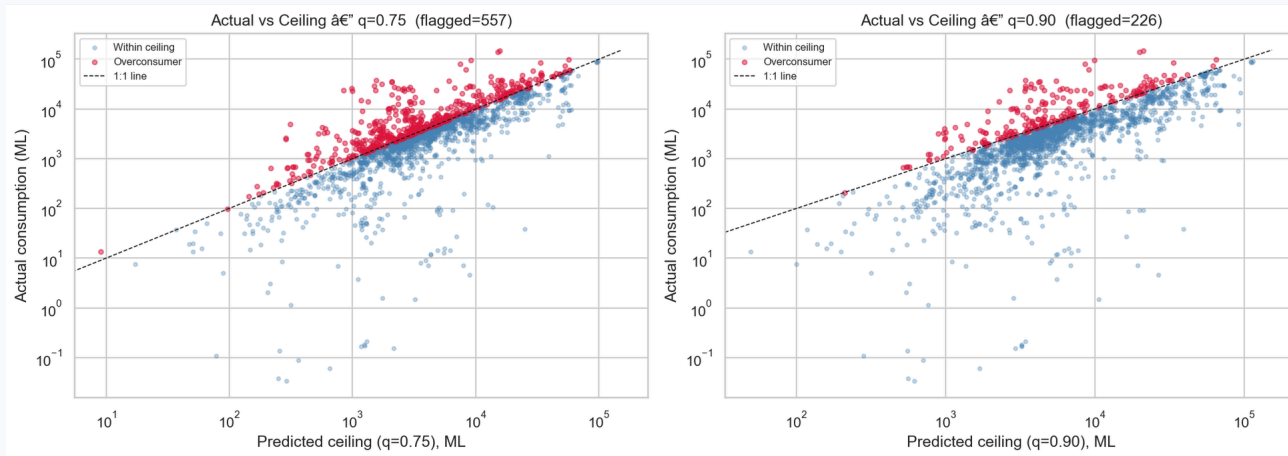
- **The Flaw of OLS:** Ordinary Least Squares minimizes squared errors to predict the conditional *mean*. In resource management, predicting the "average" plant is useless; we must identify the extreme upper-tail outliers.
- **The Solution:** We utilized **Quantile Regression ($q=0.90$)** to establish a mathematically rigorous consumption ceiling
- **Under the Hood:** Rather than squared errors, this model minimizes *asymmetric pinball loss*, heavily penalizing predictions that fall below the 90th percentile threshold for any plant's specific covariates.

What Drives Extreme Water Use?



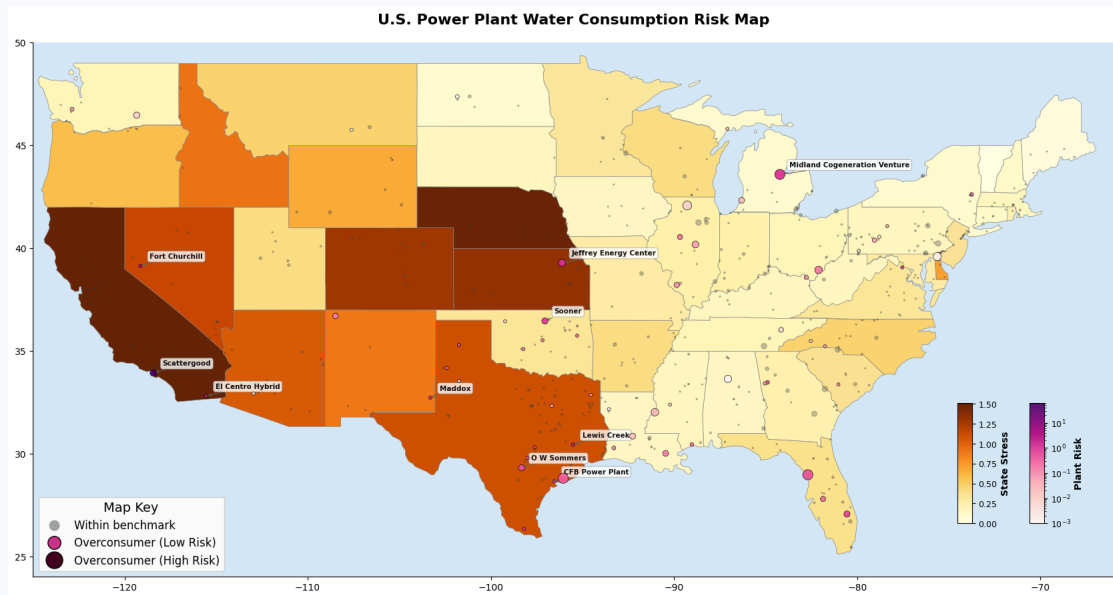
- **Key Driver:** Open-loop (once-through) cooling systems and specific fossil fuels overwhelmingly push plants into the 90th percentile.

Flagging the "Overconsumers"



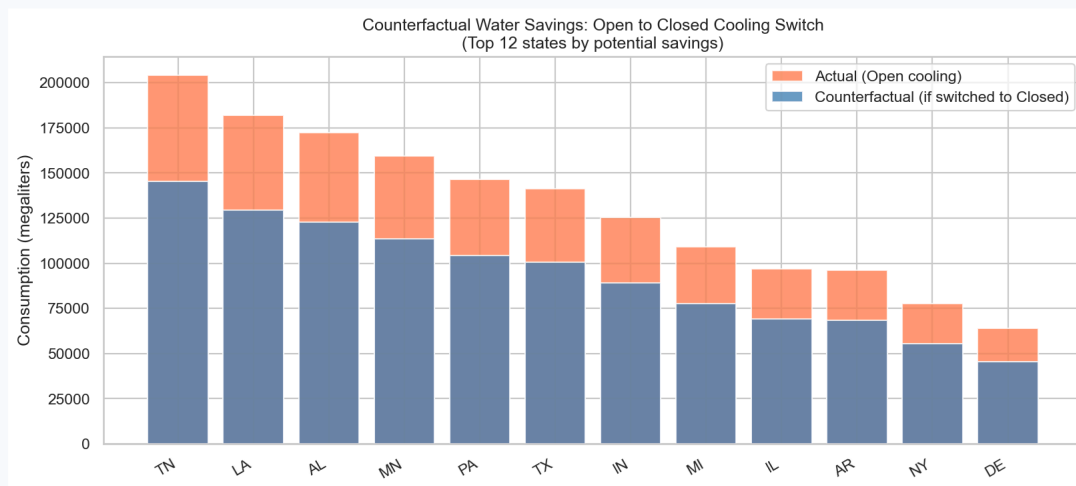
- **The Findings:** 226 plant-years (10% of dataset) plotted above the 90th percentile ceiling.
- **The Implication:** These plants operate highly inefficiently relative to their power output and peer group.

Mapping Environmental Vulnerability



- **Risk Score:**
Overconsumption
Excess $\times$ Aqueduct
Water Stress Index.
- **The Geographic Reality:** Highest risk occurs where extreme inefficiency meets drought-prone regions (Nevada, Texas, California).

The Case for Cooling Upgrades



- **The Counterfactual:** Mandating a switch from Open to Closed cooling systems.
- **The Impact:** 28.8% reduction in water use for upgraded plants.
- **National Savings:** 545,558 megaliters conserved.

Questions?

Thank You!